(except perhaps hypochondriacs). But if you enter medical school today in the expectation of still being a family doctor in twenty years, maybe you should think again. With such a Watson around, there is not much need for Sherlocks.

This threat hovers over the heads not only of general practitioners, but also of experts. Indeed, it might prove easier to replace doctors specialising in a relatively narrow field such as cancer diagnosis. For example, in a recent experiment a computer algorithm diagnosed correctly 90 per cent of lung cancer cases presented to it, while human doctors had a success rate of only 50 per cent.[8] In fact, the future is already here. CT scans and mammography tests are routinely checked by specialised algorithms, which provide doctors with a second opinion, and sometimes detect tumours that the doctors missed.[9]

A host of tough technical problems still prevent Watson and its ilk from replacing most doctors tomorrow morning. Yet these technical problems – however difficult – need only be solved once. The training of a human doctor is a complicated and expensive process that lasts years. When the process is complete, after ten years of studies and internships, all you get is one doctor. If you want two doctors, you have to repeat the entire process from scratch. In contrast, if and when you solve the technical problems hampering Watson, you will get not one, but an infinite number of doctors, available 24/7 in every corner of the world. So even if it costs $100 billion to make it work, in the long run it would be much cheaper than training human doctors.

And what's true of doctors is doubly true of pharmacists. In 2011 a pharmacy opened in San Francisco manned by a single robot. When a human comes to the pharmacy, within seconds the robot receives all of the customer's prescriptions, as well as detailed information about other medicines taken by them, and their suspected allergies. The robot makes sure the new prescriptions don't combine adversely with any other medicine or allergy, and then provides the customer with the required drug. In its first year of operation the robotic pharmacist provided 2 million prescriptions, without making a single mistake. On average, flesh-and-blood pharmacists get wrong 1.7 per cent of